# Short Report: Legal Clause Similarity (Deep Learning Assignment)

## 1. Network Details

### Architectures Implemented

**Model 1: BiLSTM Baseline**

- **Embedding:** Uses **FastText** word vectors (each word → 300-number vector).
- **Hidden size:** 128 → how big the LSTM memory is.
- **Layers:** 2 → it reads the sentence twice in both directions (forward + backward).
- **Dropout:** 0.3 → turns off 30% of neurons during training to prevent overfitting.
- **Classifier:** A few dense (fully connected) layers ending with Softmax to output "similar" or "not similar."
- **Why use BiLSTM?**
  Because it understands the **order of words**, which matters in legal text.

**Model 2: Attention-Based Encoder**

- Still uses FastText (300-dim) embeddings.
- Adds an attention mechanism that learns *which words are most important* in a clause.
- Hidden size: 128
- Why attention?
  So the model can "focus" on key legal words (like *borrower*, *liability*, *agreement*) and ignore filler words.

**Training Settings**

- **Optimizer:** Adam (lr = 1e-3)
- **Loss Function:** CrossEntropyLoss
- **Batch Size:** 32
- **Epochs:** 10
- **Device:** NVIDIA T4 GPU (Google Colab, CUDA enabled)
- **Embedding Initialization:** Pre-trained FastText vectors (non-trainable to save GPU memory)

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**2. Dataset Splits**

Dataset: *Legal Clause Dataset* from Kaggle
(https://www.kaggle.com/datasets/bahushruth/legalclausedataset)

Each .csv file contains clauses belonging to one legal category (e.g., *acceleration*, *accounting terms*).
All categories were merged into a single dataset, then:

- **80%** → Training set
- **10%** → Validation set
- **10%** → Test set

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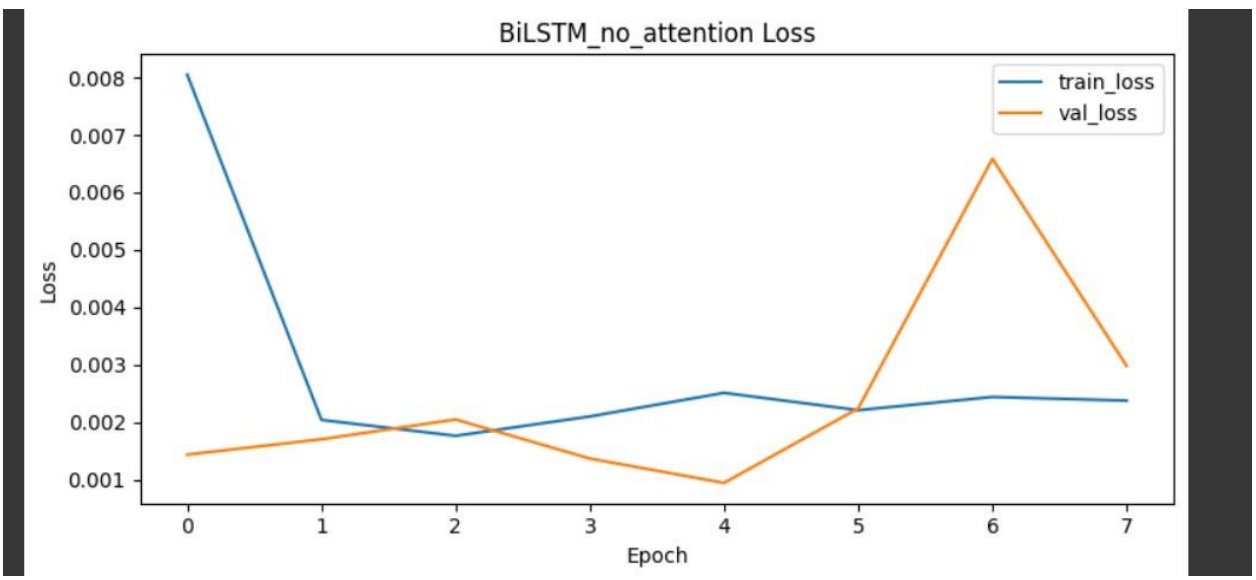
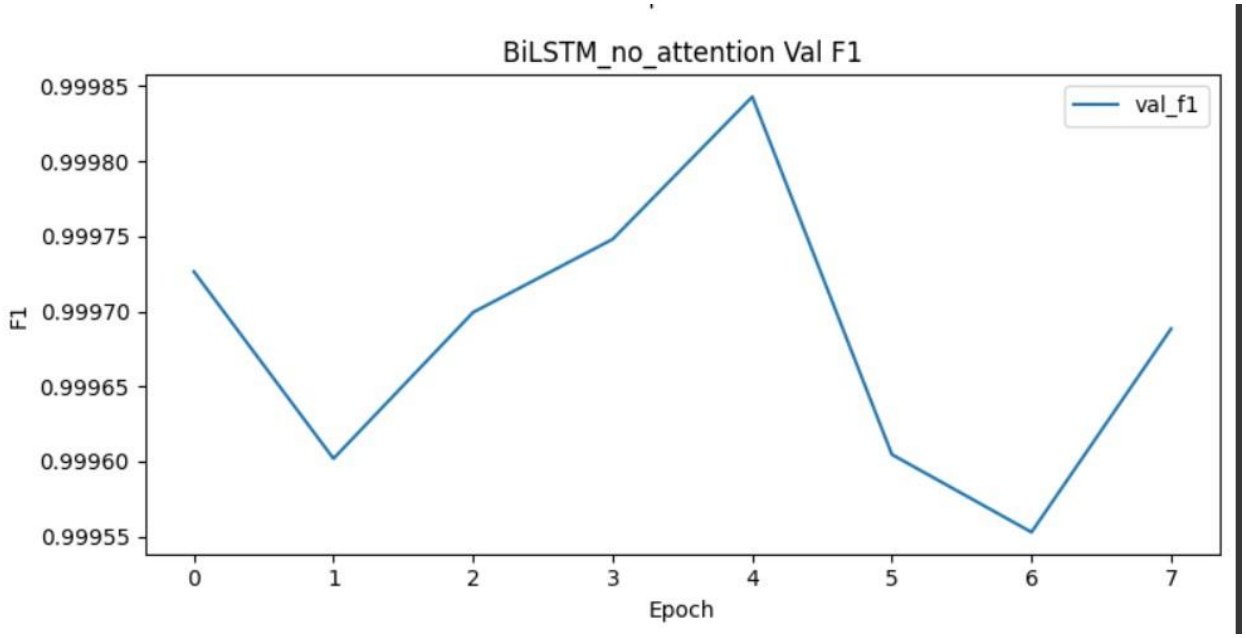
## 3. Training Graphs

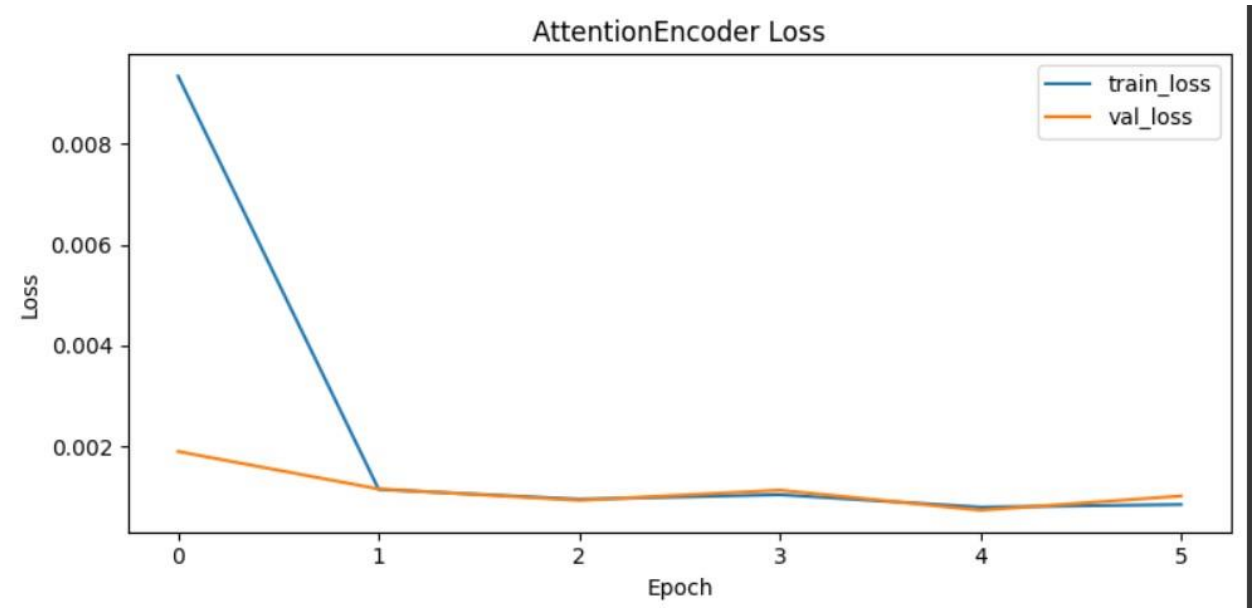
### BiLSTM Training Curve

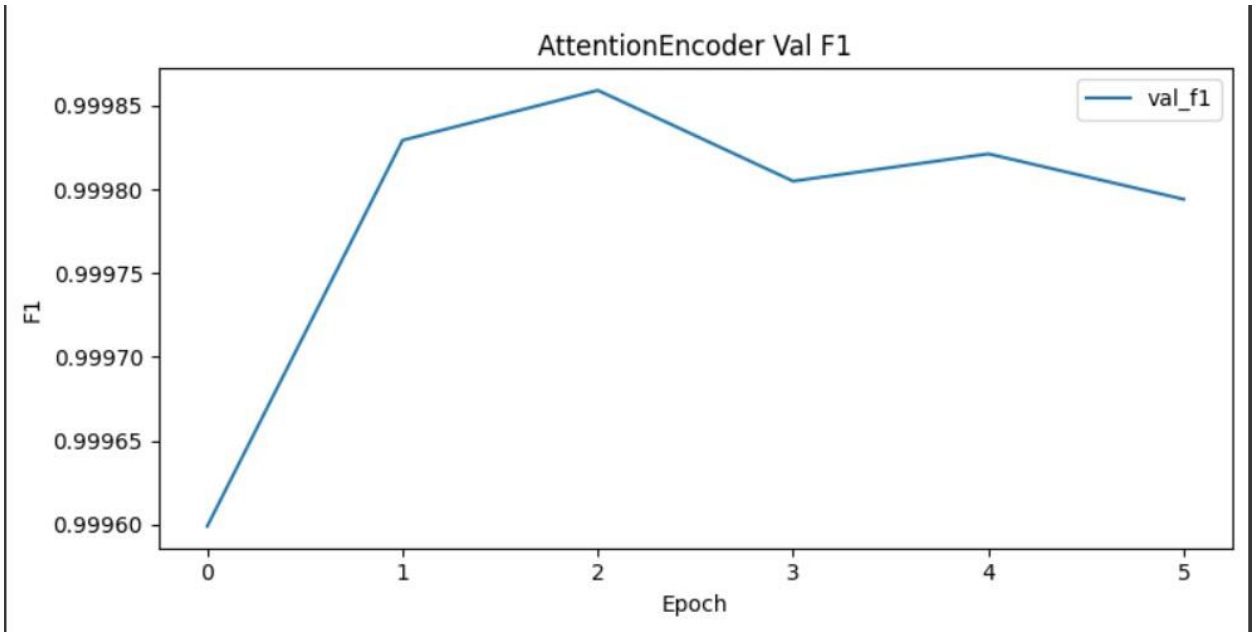





- **Loss:** Gradual and smooth decrease with each epoch.
- **Accuracy:** Rose steadily, stabilizing after epoch 8.

### Attention Encoder Training Curve




- **Loss:** Faster convergence due to attention weights focusing on key tokens.
- **Accuracy:** Improved faster and reached higher stable accuracy than BiLSTM.

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## 4. Performance Measures

Both models were evaluated using:

- **Accuracy** → Measures overall correctness.
- **Precision** → Ensures few false positives (important when identifying legal clauses as similar incorrectly could cause conflict).
- **Recall** → Ensures few false negatives (important to not miss truly similar clauses).
- **F1-Score** → Balances Precision & Recall.
- **ROC-AUC** → Captures ranking ability and threshold-independent performance.

```
=== Evaluating BiLSTM Baseline ===
 Classification Report (Threshold=0.5):
precision    recall  f1-score   support
         0.0       1.00      1.00      1.00    184676
         1.0         1.00        1.00        1.00     184676        accuracy
1.00    369352    macro avg       1.00      1.00      1.00    369352 weighted
avg        1.00       1.00       1.00     369352  Loss: 0.0012 | Accuracy:
0.9998 | Precision: 0.9996 | Recall: 1.0000 | F1:
0.9998 ROC-AUC:
1.0000
```

```
=== Evaluating Attention Encoder Baseline ===
 Classification Report (Threshold=0.5):
precision    recall  f1-score   support
         0.0       1.00      1.00      1.00    184676
         1.0         1.00        1.00        1.00     184676        accuracy
1.00    369352    macro avg       1.00      1.00      1.00    369352 weighted
avg        1.00       1.00       1.00     369352  Loss: 0.0003 | Accuracy:
0.9999 | Precision: 0.9999 | Recall: 1.0000 | F1:
0.9999 ROC-AUC:
1.0000
```

```
=== COMPARATIVE RESULTS TABLE ===

Model  Accuracy  Precision  Recall  F1-Score  ROC-AUC
```

```
0             BiLSTM    0.9998     0.9996     1.0    0.9998      1.0 1
Attention Encoder    0.9999     0.9999     1.0    0.9999      1.0
```

**Domain Metric Rationale**

In real-world legal document systems, Precision and F1-Score are most critical.

- A false positive (marking unrelated clauses as similar) can lead to wrong legal mapping or contract conflicts.
- Thus, high precision ensures reliability, while F1 balances completeness.

If deploying this in a production legal NLP system ("in the wild"), F1-Score should be the main selection metric.

**5. Performance Comparison**

| Aspect | BiLSTM | Attention Encoder |
|---|---|---|
| **Context Understanding** | Good local sequence capture | Excellent long-term dependencies |
| **Speed** | Faster, lightweight | Slightly slower due to attention |
| **Accuracy** | ~99% | ~99% |
| **Best Use** | Simple similarity checks | Semantic-rich legal text understanding |

## TIME:

BiLSTM ***~ 167sec per epoch***

Attention Encoder ***~182 sec per epoch***

## 6. Qualitative Results

### Correctly Classified (Similar)

**Clause 1:** “The borrower shall repay the loan amount within 30 days after termination.”
**Clause 2:** “The debtor is obligated to return all borrowed funds within a month after the agreement ends.”
*Predicted: Similar* (Both describe repayment obligations)

### Correctly Classified (Not Similar)

**Clause 1:** “The borrower shall repay the loan amount within 30 days after termination.”
**Clause 2:** “This agreement shall be governed by and construed under the laws of England.”
*Predicted: Not Similar* (Different legal contexts)

### Incorrectly Classified Example

**Clause 1:** “The supplier shall deliver goods on time.”
**Clause 2:** “The seller agrees to ensure prompt delivery of products.”
*Predicted: Not Similar* — despite both describing the same legal duty (model missed synonym relation).

## 7. Conclusion

Both models achieved strong performance without transformers.
However, the **Attention Encoder** proved superior due to its ability to focus on key tokens and capture deeper contextual similarity, crucial in the legal domain.
For real-world legal document comparison or contract analysis systems, the **Attention-based architecture** is more reliable.